A First Course in Abstract Algebra

Section 22: Rings of Polynomials

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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22 Rings of Polynomials

Exercise 22.1: Addition and multiplication modulo eight

Add like powers, then reduce coefficients modulo 8:

$$f + g = 2x^2 - 3 = \boxed{2x^2 + 5}.$$

For the product, distribute before reducing:

$$fg = 8x^3 - 26x^2 + 28x - 10 = \boxed{6x^2 + 4x + 6} \quad \text{in } \mathbb{Z}_8[x].$$

The degree drops because the leading coefficient 8 vanishes modulo 8.

Exercise 22.2: Addition and multiplication in characteristic two

Here $f = g = x+1$. Thus $f+g = 2x+2 = \boxed{0}$. Expanding the product gives $fg = x^2+2x+1 = \boxed{x^2+1}$, since $2 = 0$ in $\mathbb{Z}_2$.

Exercise 22.3: Polynomial arithmetic modulo six

Addition gives

$$f + g = 5x^2 + 5x + 7 = \boxed{5x^2 + 5x + 1}.$$

Distributive multiplication over the integers gives

$$fg = 6x^4 + 13x^3 + 24x^2 + 17x + 12.$$

Reducing each coefficient modulo 6 gives $\boxed{x^3 + 5x}$. The expected degree 4 disappears because $2 \cdot 3 = 0$ in the coefficient ring.

Exercise 22.4: Polynomial arithmetic modulo five

After collecting terms,

$$f + g = 3x^4 + 2x^3 + 4x^2 + 5x + 6 = \boxed{3x^4 + 2x^3 + 4x^2 + 1}.$$

The unreduced product is

$$6x^7 + 12x^6 + 9x^5 + 10x^4 + 16x^3 + 22x^2 + 16x + 8.$$

Thus in $\mathbb{Z}_5[x]$,

$$\boxed{fg = x^7 + 2x^6 + 4x^5 + x^3 + 2x^2 + x + 3}.$$

Exercise 22.5: Counting short binary polynomials

Every polynomial of degree at most 3, including zero, is uniquely $a_0 + a_1x + a_2x^2 + a_3x^3$. Each coefficient has two choices in $\mathbb{Z}_2$, independently. The count is $\boxed{2^4 = 16}$. Allowing leading coefficients zero includes all smaller degrees.

Exercise 22.6: Counting short polynomials modulo five

Write the polynomial uniquely as $a_0 + a_1x + a_2x^2$, allowing leading zeros. Each of the three coefficients has five choices. The answer is $\boxed{5^3 = 125}$, including the zero polynomial.

Exercise 22.7: Evaluation at a real point

Evaluation means substituting the indicated input. Therefore $\phi_2(x^2 + 3) = 2^2 + 3 = \boxed{7}$.

Exercise 22.8: Evaluation at the imaginary unit

Substitute i and use $i^2 = -1$, $i^3 = -i$:

$$\phi_i(2x^3 - x^2 + 3x + 2) = 2(-i) - (-1) + 3i + 2 = \boxed{3 + i}.$$

Exercise 22.9: Evaluation of a product modulo seven

Evaluation preserves products, so evaluate the two factors separately at 3. They give $3^4 + 2 \cdot 3 = 87 \equiv 3$ and $3^3 - 3 \cdot 3^2 + 3 = 3$. Their product is $9 \equiv \boxed{2} \pmod 7$.

Exercise 22.10: Three evaluated factors modulo seven

At $x = 5$ the factors reduce modulo 7 to

$$5^2 + 2 = 27 \equiv 6, \quad 4 \cdot 5^2 + 3 = 103 \equiv 5, \quad 5^7 + 3 \cdot 5^2 + 1 \equiv 5 + 5 + 1 \equiv 4.$$

Here $5^7 \equiv 5$ by Fermat. Multiply the residues: $6 \cdot 5 \cdot 4 = 120 \equiv \boxed{1} \pmod 7$.

Exercise 22.11: Large exponents evaluated modulo seven

Since $4 \neq 0$ in $\mathbb{Z}_7$, Fermat gives $4^6 = 1$. Reduce the exponents: $106 \equiv 4$, $99 \equiv 3$, $53 \equiv 5 \pmod 6$. The residues $4^3 = 1$, $4^4 = 4$, $4^5 = 2$ in $\mathbb{Z}_7$ give

$$\phi_4(3x^{106} + 5x^{99} + 2x^{53}) = 3 \cdot 4 + 5 \cdot 1 + 2 \cdot 2 = 21 = \boxed{0}.$$

Exercise 22.12: Zeros over two elements

There are only two inputs. At 0, $x^2 + 1 = 1$; at 1, it is $1 + 1 = 0$ in $\mathbb{Z}_2$. Thus the complete zero set is $\boxed{\{1\}}$.

Exercise 22.13: Zeros of a cubic over seven elements

Evaluating at $0, 1, 2, 3, 4, 5, 6$ gives, respectively, residues $2, 5, 0, 0, 4, 4, 6$. Hence the zeros are exactly $\boxed{2, 3}$. In particular, a cubic may have a repeated root rather than three distinct roots.

Exercise 22.14: Zeros of a fifth-degree polynomial over five elements

For each $a \in \mathbb{Z}_5$, Fermat gives $a^5 = a$, so the value becomes $3a^3 + a^2 + 3a$. At $a = 0, 1, 2, 3, 4$ the residues are $0, 2, 4, 4, 0$. Therefore the complete zero set is $\boxed{\{0, 4\}}$.

Exercise 22.15: Zeros of a product

Since $\mathbb{Z}_7$ is a field, $f(a)g(a) = 0$ if and only if at least one factor is zero. Evaluating $f = x^3 + 2x^2 + 5$ at $0, \ldots, 6$ gives $5, 1, 0, 1, 3, 5, 6$, so its only zero is 2. Factoring $g = 3x^2 + 2x = x(3x + 2)$ gives zeros 0 and 4, because $3 \cdot 4 + 2 = 14 = 0$. Taking the union, the product has zeros $\boxed{0, 2, 4}$.

Exercise 22.16: Evaluation with prime-field exponent reduction

For nonzero $3 \in \mathbb{Z}_5$, $3^4 = 1$. The exponents satisfy $231 \equiv 3$, $117 \equiv 1$, $53 \equiv 1 \pmod 4$. Hence

$$\phi_3(x^{231} + 3x^{117} - 2x^{53} + 1) = 3^3 + 3 \cdot 3 - 2 \cdot 3 + 1 = 31 \equiv \boxed{1} \pmod 5.$$

Exercise 22.17: All zeros of a large-degree polynomial

At $x = 0$ the polynomial is zero, since every exponent is positive. For $x \neq 0$, use $x^4 = 1$ in $\mathbb{Z}_5$ and reduce exponents modulo 4:

$$2x^{219} + 3x^{74} + 2x^{57} + 3x^{44} = 2x^3 + 3x^2 + 2x + 3.$$

This cubic factors as $(x^2 + 1)(2x + 3)$. Its zeros are $x = 2, 3$ from $x^2 = -1 = 4$, and $x = 1$ from $2x = -3 = 2$. Including the separately checked zero input, all zeros are $\boxed{0, 1, 2, 3}$. The exponent reduction was not applied blindly to $x = 0$.

Exercise 22.18: Correcting the definition of a polynomial

Add the condition that only finitely many coefficients are nonzero. A polynomial over R is a formal expression $\sum_{i \geq 0} a_i x^i$ with $a_i \in R$ and $a_i = 0$ for all sufficiently large i. Without that finiteness condition, the expression is a formal power series rather than necessarily a polynomial.

Exercise 22.19: The definition of a zero

The stated definition is correct: $a \in F$ is a zero of f when evaluation at a gives $\phi_a(f) = f(a) = 0$. More generally, if $F \subseteq E$, a zero in E is an element $a \in E$ satisfying the same equation under $\phi_a : F[x] \to E$. The domain of the evaluation map is the polynomial ring $F[x]$.

Exercise 22.20: Regrouping a polynomial in two variables

Collect the displayed expression by powers of x rather than powers of y. The x^4 terms contribute $(y+1)x^4$; the x^3 term is $3y^3x^3$; the x^2 terms give $(y^2-3)x^2$; and the linear terms give $(2y^3-6y^2-2y)x$. The remaining constant in x is $y^2 + 2$. Thus in $(\mathbb{Q}[y])[x]$ it is

$$\boxed{(y + 1)x^4 + 3y^3x^3 + (y^2 - 3)x^2 + (2y^3 - 6y^2 - 2y)x + (y^2 + 2)}.$$

Exercise 22.21: Six kernel elements for evaluation at five

The kernel consists of polynomials with rational coefficients vanishing at 5. Six distinct examples are

$$0, \quad x-5, \quad 2(x-5), \quad (x-5)^2, \quad x(x-5), \quad (x-5)^3.$$

Every displayed expression has value zero after substituting $x = 5$, which directly verifies membership. Their different coefficients or degrees show that the six polynomials are distinct.

Exercise 22.22: A positive-degree polynomial unit

In $\mathbb{Z}_4[x]$, take $u = 1 + 2x$. Its square is $u^2 = 1 + 4x + 4x^2 = 1$. Therefore u is a unit, with inverse itself, and has degree $1 > 0$. This phenomenon uses the zero-divisor coefficient 2.

Exercise 22.23: True or false: polynomial rings

(a) True: a formal polynomial is zero exactly when each coefficient is zero.

(b) True: commutative coefficient products allow the convolution formula for multiplication to be reversed.

(c) True: the product of nonzero leading coefficients in a domain remains nonzero, so nonzero polynomials cannot multiply to zero.

(d) True: regard a pair of nonzero zero-divisor coefficients as constant polynomials.

(e) False: multiplication cannot create a term of degree greater than $3 + 4 = 7$.

(f) False: in $\mathbb{Z}_4[x]$, $(2x^3)(2x^4) = 0$, so the degree need not be 7.

(g) True: evaluation is multiplicative, giving $(fg)(a) = f(a)g(a) = 0$.

(h) True: if $fg = 1$ over a field, degree additivity forces both degrees to be zero; the units are nonzero constants.

(i) True whenever x is interpreted as the polynomial indeterminate in a ring with unity: multiplication by x shifts each coefficient one place, so $xf = 0$ forces $f = 0$. The same coefficient-shift statement holds formally without a unity.

(j) False: $2x$ is a nonconstant zero divisor in $\mathbb{Z}_4[x]$, since $(2x)2 = 0$.

Exercise 22.24: A polynomial ring over a domain is a domain

The standard coefficient operations give a commutative ring $D[x]$ with unity the constant polynomial $1_D \neq 0$. Let f, g be nonzero, with leading terms $a_m x^m, b_n x^n$. The coefficient of x^{m+n} in fg is $a_m b_n$, which is nonzero because D has no zero divisors. Hence $fg \neq 0$. Thus $D[x]$ has no nonzero zero divisors and is an integral domain.

Exercise 22.25: Units in three polynomial rings

(a) If $fg = 1$ in $D[x]$, then $\deg f + \deg g = \deg 1 = 0$ by the leading-coefficient argument. Both degrees must be zero, so f is a constant whose value is a unit in D. Conversely every such constant has its constant inverse in $D[x]$.

(b) The units of $\mathbb{Z}$ are ± 1, so the units of $\mathbb{Z}[x]$ are precisely the two constant polynomials $1, -1$.

(c) Every nonzero element of the field $\mathbb{Z}_7$ is a unit. Thus the units of $\mathbb{Z}_7[x]$ are precisely the constant polynomials $1, 2, 3, 4, 5, 6$.

Exercise 22.26: Left distributivity from coefficient formulas

Write $f = \sum a_i x^i$, $g = \sum b_j x^j$, and $h = \sum c_j x^j$, extending coefficient lists by zeros. The coefficient of x^n in $f(g + h)$ is

$$\sum_{i+j=n} a_i(b_j + c_j) = \sum_{i+j=n} a_i b_j + \sum_{i+j=n} a_i c_j.$$

Distributivity in R gives the equality and its abelian addition allows regrouping the finite sum. The right side is exactly the coefficient of x^n in $fg + fh$. Equality for every coefficient proves $f(g + h) = fg + fh$.

Exercise 22.27: Formal differentiation as an additive map

(a) Coefficientwise differentiation gives $D(f+g) = Df+Dg$, so D is an additive-group homomorphism. It is not a ring homomorphism: $D(x^2) = 2x$ but $D(x)D(x) = 1$. These formal polynomials are distinct.

(b) If $Df = 0$, every coefficient ka_k with $k \geq 1$ vanishes. In characteristic zero the scalar k is nonzero and invertible in the field, so $a_k = 0$ for $k \geq 1$. Hence the kernel is exactly the constant polynomials F.

(c) For $g = \sum_{k=0}^{n} b_k x^k$, set $f = \sum_{k=0}^{n} (b_k/(k+1)) x^{k+1}$. All denominators are nonzero in the characteristic-zero field. Then $Df = g$, so the image is all of $F[x]$.

Exercise 22.28: Evaluation in several variables

(a) For $\boldsymbol{a} = (a_1, \ldots, a_n) \in E^n$, define

$$\phi_{\boldsymbol{a}}\left(\sum_{\boldsymbol{k}} c_{\boldsymbol{k}} x_1^{k_1} \cdots x_n^{k_n}\right) = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}} a_1^{k_1} \cdots a_n^{k_n}.$$

All sums are finite. Distributing products in the commutative field E proves that this map preserves addition and multiplication. It fixes the coefficients in F and sends x_i to a_i.

(b) Substitution yields $(-3)^2 2^3 + 3(-3)^4 2 = 9 \cdot 8 + 3 \cdot 81 \cdot 2 = \boxed{558}$.

(c) A zero in E^n is a tuple $\boldsymbol{a}$ with $\phi_{\boldsymbol{a}}(f) = f(a_1, \ldots, a_n) = 0$. A tuple, rather than a single scalar, is required.

Exercise 22.29: Pointwise operations on all functions

The zero function is the additive identity, and $(-\phi)(r) = -\phi(r)$ provides additive inverses. Addition is commutative and associative because these identities hold at every value r in R. For multiplication,

$$((\phi\psi)\theta)(r) = (\phi(r)\psi(r))\theta(r) = \phi(r)(\psi(r)\theta(r)) = (\phi(\psi\theta))(r).$$

Likewise evaluating either distributive law at r gives the corresponding law in R, for example

$$(\phi(\psi + \theta))(r) = \phi(r)(\psi(r) + \theta(r)) = (\phi\psi + \phi\theta)(r).$$

Closure is automatic because the operations in R have values in R. Thus the function set is a ring. Its multiplication is pointwise multiplication, as stipulated, not composition.

Exercise 22.30: Polynomial functions form a smaller kind of ring

(a) The zero function is polynomial. If functions u, v are represented by polynomials f, g, then $u - v$ is represented by $f - g$, and uv by fg. The subring test proves that P_F is a subring of the full function ring.

(b) If F has q elements, there are only q^q functions $F \to F$, hence only finitely many polynomial functions. But the formal polynomials $1, x, x^2, \ldots$ are infinitely many distinct members of $F[x]$. A bijection, and thus a ring isomorphism, is impossible. Distinct formal polynomials can induce the same function; over $\mathbb{Z}_p$, for instance, x^p and x do so.

Exercise 22.31: All functions on a finite field are polynomial

(a) Each input can be assigned any output independently. There are $\boxed{2^2 = 4}$ functions $\mathbb{Z}_2 \to \mathbb{Z}_2$, and $\boxed{3^3 = 27}$ functions $\mathbb{Z}_3 \to \mathbb{Z}_3$.

(b) Recording the two, or three, function values is an additive-group isomorphism to a Cartesian power. Thus the two groups are $\boxed{\mathbb{Z}_2 \times \mathbb{Z}_2}$ and $\boxed{\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3}$ respectively. Addition in each coordinate is the field addition.

(c) List $F = \{a_1, \ldots, a_q\}$. Given any function $u : F \to F$, define

$$f(x) = \sum_{i=1}^{q} u(a_i) \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}.$$

Every denominator is nonzero because the a_i are distinct. At $x = a_k$, all summands but the kth contain a zero factor; the kth product equals 1. Therefore $f(a_k) = u(a_k)$ for every k. This polynomial represents u, proving $F^F = P_F$.